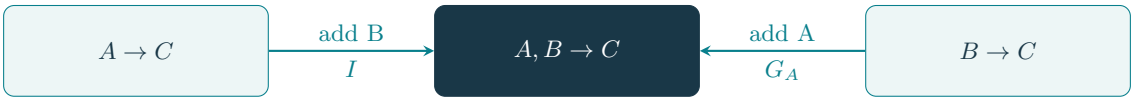
# Complex spatial rules

What to keep, what to compare, and what the tests establish

**Recommendation.** Keep lift and your $I$. For $A, B \to C$, also check the gain over $B \to C$ and test that second gain. A large $I$ alone can be entirely explained by $B \to C$.

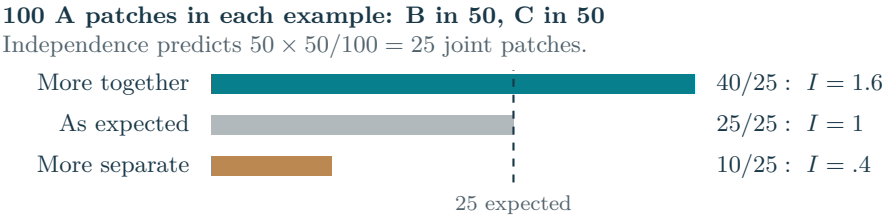
## 1 One population, two contributions

$A$ means **center type A**; $B, C$ mean **distinct neighbor types present**. Count each retained patch once, within one sample. All unconditioned counts use the same patches across *all* center types. Write $n_{ABC}$ for patches containing all three events, and similarly for other counts.




$$I = \frac{P(C \mid AB)}{P(C \mid A)} = \frac{P(BC \mid A)}{P(B \mid A)P(C \mid A)} = \frac{n_{ABC}n_A}{n_{AB}n_{AC}},$$
$$G_A = \frac{P(C \mid AB)}{P(C \mid B)} = \frac{n_{ABC}n_B}{n_{AB}n_{BC}}.$$

**$I$ asks whether B adds information beyond A; $G_A$ asks whether A adds information beyond B.** A gain of 1.2 is a 20% relative increase, not 20 percentage points. $I$ is symmetric in B and C; the prediction rule and $G_A$ are directional.




## 2 Recommended gates

The two rule lifts use different baselines:

$$L_J = \frac{P(BC \mid A)}{P(BC)}, \qquad L_K = \frac{P(C \mid AB)}{P(C)}.$$

| Positive rule | Effect criteria | Meaning in the observed data |
|---|---|---|
| $A \to B, C$ | $L_J \geq 1.2$, $I \geq 1.2$ | BC is enriched around A and exceeds the product of its two frequencies within A. |
| $A, B \to C$ | $L_K \geq 1.2$, $I \geq 1.2$, $G_A \geq 1.2$ | C is enriched and prediction improves over either antecedent alone. |

**Significance:** retain the rule's own permutation $q_{\text{rule}} \leq .05$. Also require $q_I \leq .05$ for $A \to B, C$, or the combined $q_{\text{add}} \leq .05$ for $A, B \to C$ (page 2).

**Negative rules:** replace each effect threshold by $\leq 1/1.2 \approx .833$; retain the significance gates. Complex antecedents must improve avoidance over *both* simpler predictions. A negative joint rule describes fewer BC combinations, not necessarily avoidance of each neighbor individually.

**Keep roles explicit.** Here $B \to C$ means B-neighbor presence predicting C-neighbor presence in the original patches. Your code's B-center→C-neighbor comparator changes the patch geometry. It answers a different biological question.

These criteria support the descriptive claims above. They do not establish causation or a B–C interaction stronger around A than elsewhere. Spatial calibration governs the statistical confidence, not the arithmetic.

## 3 Apply the same Fisher test twice

**B adds beyond A**

Restrict to A-centered patches.

| | C | No C |
|---|---|---|
| B | $n_{ABC}$ | $n_{AB} - n_{ABC}$ |
| No B | $n_{AC} - n_{ABC}$ | $n_A - n_{AB} - n_{AC} + n_{ABC}$ |

$p_B$: Fisher test of $I = 1$.

**A adds beyond B**

Restrict to B-containing patches.

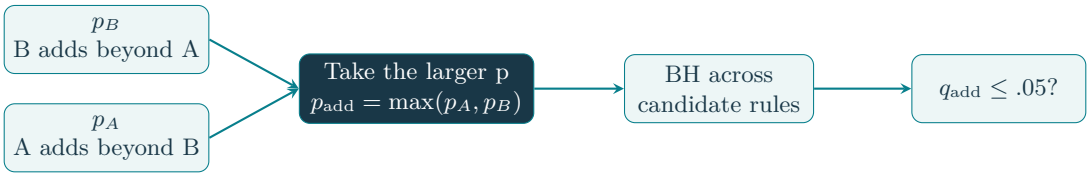
| | C | No C |
|---|---|---|
| A | $n_{ABC}$ | $n_{AB} - n_{ABC}$ |
| Not A | $n_{BC} - n_{ABC}$ | $n_B - n_{AB} - n_{BC} + n_{ABC}$ |

$p_A$: Fisher test of $G_A = 1$.

Use two-sided Fisher tests, then require the desired effect direction. Fisher tests odds ratio 1; the odds ratio itself is *not* $I$ or $G_A$. For any table $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, with $N = a + b + c + d$ and fixed margins:

$$h(x) = \frac{\binom{a+c}{x}\binom{b+d}{a+b-x}}{\binom{N}{a+b}}, \qquad p = \sum_{x:\, h(x) \le h(a)} h(x).$$

**For complex antecedents, the weaker comparison decides:**

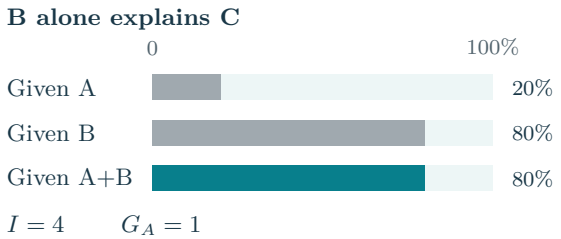



BH sorts the $m$ raw combined p-values and uses $q_{(i)} = \min\{1, \min_{j \ge i} mp_{(j)}/j\}$. Both comparisons must succeed. If either independence null is true, a small maximum requires its p-value to be small too. This combination is valid when its component tests are valid, even if they depend on each other.

Apply BH to all planned complex-antecedent candidates per sample, including attraction and avoidance, before selecting favorable results. For joint rules, correct $p_B$ separately across centered triplets with unordered {B,C}, giving $q_I$. Missing comparators must be measured from the patches; missing or untestable values cannot pass.

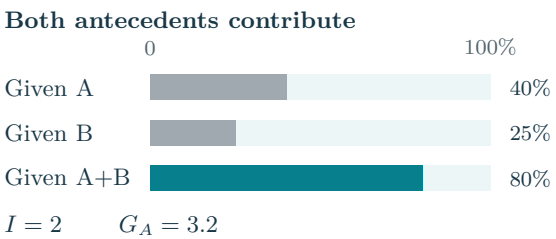
## 4 The pitfall and a rule that survives it

**How often is C present?** Both panels use the same 0–100% scale.




$I = 4$ $G_A = 1$

Adding A changes nothing.




$I = 2$ $G_A = 3.2$

The pair improves on both.

Both examples have 100 A patches among 1,000 total patches. Counts are $(n_{AB}, n_{AC}, n_{ABC})$; $(n_B, n_C, n_{BC})$; both have $L_K = 4$.

| | **B alone explains C** | **Both antecedents contribute** |
|---|---|---|
| Counts | $(20, 20, 16)$; $(200, 200, 160)$ | $(20, 40, 16)$; $(200, 200, 50)$ |
| Fisher $p_B$, $p_A$ | $1.45 \times 10^{-11}$, 1 | $7.11 \times 10^{-5}$, $6.53 \times 10^{-8}$ |
| $p_{\text{add}}$ | 1: reject as an added-value rule | $7.11 \times 10^{-5}$: eligible for BH |

Single-condition bars include patches containing both A and B. Right: $.8/.4 = 2$ and $.8/.25 = 3.2$. Acceptance also needs the full-family correction and the spatial permutation result.

## 5 Two signs can describe different things

**Mixed signs:** in 100 A patches, B and C each occur in 20, jointly in 10. If global $P(BC) = .20$, then $L_J = .5$ but $I = 2.5$: the pair is depleted around A, yet associates within A. **Negative redundancy:** B and C frequencies .5 and joint frequency .1 in both populations give $I = L_K = .4$, but $G_A = 1$; B alone still explains the avoidance.

## 6 What changes from your current code

| | Current behavior | Recommended interpretation |
|---|---|---|
| Complex antecedents | Lift beats qualifying simpler rules; default gain 1.1; roles stripped for matching. | Compare both gains in the same patch population; your proposed threshold is 1.2. |
| Complex consequents | Conviction beats qualifying simpler rules; default gain 1.1. | Use I for the within-A co-occurrence claim. |
| No significant simpler | Marked as adding information. | Still evaluate both simpler explanations; no qualifying comparator is not evidence of improvement. |
| Conditional test | Freeze simpler-rule cell types; shuffle other labels; compare observed gain. | The proposed Fisher tables test different nulls. Reuse the maximum-p aggregation, not the old p-values. |

**Why drop conviction here?** Conviction is $(1 - P(Y))/(1 - P(Y \mid X))$; changing Y changes the failure event and baseline. For the 40-joint-patch example, global $P(B) = P(C) = .25$ and $P(BC) = .10$ give single and joint convictions of 1.5, despite $I = 1.6$.

**What your permutation q means:** the raw score is

$$p_{\text{rule}} = \frac{1 + \#\{\text{shuffles passing the rule's mining thresholds}\}}{M + 1}.$$

It measures threshold-survival rarity, not how often observed lift is exceeded. The optional conditional shuffle instead compares observed gains. If dropping conviction, also disable its mining/shuffle threshold when configured.

## 7 Design decisions to settle

| Question | Tradeoff and suggested choice |
|---|---|
| Must both A and B contribute? | **Yes** for a nonredundant $A, B \to C$ claim: both gain thresholds and $q_{\text{add}}$. Decide separately whether B-centered rules should also block reporting. |
| Must association be stronger around A than elsewhere? | This is a stronger target than the proposed descriptive rules. **Still open.** Even positive joint lift and I do not establish it. |
| Must a joint rule have a significant individual neighbor? | Keep your gate if that defines your category. It excludes joint-only enrichment: e.g. each neighbor occurs in 20% both within A and globally, but BC occurs in 16% within A versus 5% globally. |
| Does nonsignificant A→B define context? | It can define a reporting label: "no detected A→B enrichment." It does not establish neutrality. Unknown q-values cannot satisfy it. |
| Show both rule phrasings or choose one? | If A→C is significant and A→B is not, both original definitions may pass. Show both, or choose a display priority. This choice does not establish redundancy. |
| Which negative patterns matter? | Requiring low I selects additional separation and excludes mixed signs. Low joint lift describes fewer BC combinations; individual avoidance needs individual evidence. |

## 8 Statistical limits, kept in proportion

**Shared cells make patches dependent;** Fisher may overstate evidence. Patch size and crowding filters can induce association. Existing shuffles retain geometry but assume exchangeable labels and test a different null. Calibration remains unresolved.

**BH needs valid p-values and suitable dependence across tests.** Combining gates and later filtering does not automatically establish 5% FDR for the final categories. Significance against 1 plus an estimated gain of 1.2 does not prove the population gain exceeds 1.2. Undefined gains and constant margins cannot establish added value.

Thresholds are design choices: 1.2 and .833 mean +20% and −16.7%. Also $I \leq 1/\max\{P(B \mid A), P(C \mid A)\}$; 90% neighbor prevalence makes $I \geq 1.2$ impossible.

**References:** Hämäläinen & Webb (2019), §4.2: conditional improvement; SciPy: Fisher calculation; SciPy: BH assumptions; Phipson & Smyth (2010): permutation p-values.

**Code:** `complex_rules.py`, `rules.py`, `validation/*.py`. Examples verified; policy not implemented.